

# THE CATASTROPHE LADDER OF NOISE-INDUCED ESCAPE: FROM TRACY–WIDOM TO RESURGENCE

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**ABSTRACT.** Noise-induced escape at a slow–fast catastrophe singularity is governed by one reusable pipeline: geometric blow-up, a Cole–Hopf/Riccati linearization, a stochastic edge operator, an edge law, and a resurgent trans-series. Instantiated at the fold ( $A_2$ ) the edge law is exactly Tracy–Widom; at the cusp ( $A_3$ ) it is a new, provably non-integrable law  $\mathcal{W}$  whose only closed form is a resurgent trans-series; and the whole family is organized by the turning order  $q$  into a ladder of *swept multicritical* edge laws with Bessel-1/( $q+2$ ) skeletons and exact Freidlin–Wentzell tails. We collect the proved spine, the resurgent structure and its Stokes data, a leading-order edge-variance formula uniform across the ladder, and a sharp separation between this swept ladder and the genuine (Berry–Upstill) diffraction-catastrophe ladder. Throughout we distinguish what is proved, what is numerical, and where the program crosses into a neighbouring field; provenance is tagged on every claim.

## 1. INTRODUCTION: ONE FAMILY, ONE PIPELINE

This program occupies one specific intersection: *noise-induced escape at slow–fast catastrophe singularities*, studied through a single characterization pipeline,

geometric blow-up  $\rightarrow$  Cole–Hopf/Riccati  $\rightarrow$  stochastic edge operator  $\rightarrow$  edge law  $\rightarrow$  resurgence.

That pipeline is the program’s identity, and this paper is its synthesis. Its purpose is to state, in one place and with honest provenance, what has been proved, what is established only numerically, and where the natural questions cross out of the subject into random matrix theory, resurgence proper, integrable probability, or computational neuroscience.

The organizing parameter is the turning order  $q$  of the swept inner equation, indexing the  $A_{q+1}$  singularity:  $q=1$  fold ( $A_2$ ),  $q=2$  cusp ( $A_3$ ),  $q=3$  swallowtail ( $A_4$ ), and so on. The fold rung is rigorous—the escape law is exactly Tracy–Widom (§3)—and carries a full resurgent structure with an exact Stokes constant (§4). The cusp rung is a genuinely new, *provably non-integrable* law  $\mathcal{W}$  (§5), whose only viable closed form is a resurgent trans-series. Above the cusp the family is a ladder of swept multicritical edge laws (§6); we give a leading-order edge-variance formula valid on every rung, verify the swallowtail rung, and separate this swept ladder sharply from the genuine higher diffraction catastrophes (Pearcey, swallowtail function), which form a distinct family. Section 7 anchors the theory in the coupled neuron model—honestly, as a consolidation within the Berglund–Gentz–Kuehn / Freidlin–Wentzell framework, not as new phenomena—and §8 demarcates the open program.

A word on the two standards of rigour we invoke. “Proved” means a theorem, sometimes modulo a cited black box (Ramírez–Rider–Virág for the stochastic Airy operator; Costin for the summability of the Painlevé II trans-series). “Numerical” means a high-precision computation, often exact in the sense of Monte-Carlo-free Wiener-chaos evaluation, but not a proof. We never declare a theorem where a gap remains.

## 2. THE UNIVERSAL PIPELINE

We state the machinery once, abstractly; each later section is an instantiation.

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**1. Normal form.** Near a slow-fast fold the reduced dynamics is  $dr = (b(\theta)r^2 - a(\theta)y) dt + \sigma dW$ ,  $dy = -\varepsilon_2 c(\theta) dt$ , with  $a, b, c > 0$  smooth on the slow circle. The deterministic part is the standard folded normal form; the additive noise is the added ingredient.

**2. Geometric blow-up.** The Krupa–Szmolyan quasi-homogeneous blow-up [1, 2] with weights  $(1, 2, 3)$  on  $(r, y, \varepsilon_2)$  resolves the fold; the central rescaling chart  $K_2$  ( $\rho = \varepsilon_2^{1/3}$ ) carries the inner dynamics.

**3. Canonical inner Riccati.** In  $K_2$  the SDE becomes  $dR = (R^2 - Y) dT + \eta dB$ ,  $Y = Y_0 - T$ , with effective noise  $\eta = \sigma/\sqrt{\varepsilon_2}$  (an exact Itô change of variables). The deterministic inner solution is the canard  $R = \text{Ai}'(Y)/\text{Ai}(Y)$ , escaping at the first Airy zero  $a_1 \approx -2.3381$ .

**4. Cole–Hopf linearization.** The substitution  $R = -u'/u$  turns the Riccati into a *linear* stochastic Schrödinger equation  $u'' = (Y - \eta\xi)u$ ,  $\xi = dB/dT$ , with *no* Itô–Stratonovich correction; a finite-time blow-up  $R \rightarrow +\infty$  is exactly a simple zero (node) of  $u$ . Escape becomes the first node  $Y^*$  swept downward.

**5. Stochastic edge operator.** Writing  $x = T$ , this is the eigenvalue equation for the stochastic Airy operator  $H_\beta = -\partial_x^2 + x + \frac{2}{\sqrt{\beta}}b'(x)$  on  $[0, \infty)$  with Dirichlet boundary condition [3], under the dictionary  $\eta = 2/\sqrt{\beta}$ .

**6. Edge law.** A shooting characterization identifies the *swept* first-node law with the *fixed* half-line operator’s ground state, via the monotone Dirichlet ground-state function; because the sweep level is deterministic, white-noise stationarity gives  $Y^* =_d -\Lambda(\beta) =_d \text{TW}_\beta$ .

The claim of the pipeline is a universality statement: any slow passage whose inner equation is the Airy/Riccati canard, perturbed by white noise, Cole–Hopf-linearizes to a Schrödinger operator with white-noise potential, hence to the stochastic Airy operator and the same  $\text{TW}_\beta$  exit law, with  $\beta$  fixed by the noise-to-curvature ratio. Replacing the fold’s linear turning potential  $Y$  by the multicritical  $\text{sign}(Y)|Y|^q$  generates the whole ladder of §6.

### 3. THE PROVED RUNG: FOLD $\rightarrow$ TRACY–WIDOM

**Theorem 1** (Fold escape is Tracy–Widom; RRV cited). *For the inner Riccati with  $\eta = 2/\sqrt{\beta}$ , the peel-off level satisfies  $Y^* =_d -\Lambda(\beta) =_d \text{TW}_\beta$ , where  $\Lambda(\beta)$  is the ground-state eigenvalue of  $H_\beta$ . Consequently the early-escape probability is  $1 - F_\beta(0)$  and, as  $\eta \rightarrow 0$ , the escape rate is  $\exp(-c/\eta^2)$  with  $c$  a Painlevé II action. [proved elsewhere, cited]*

Here  $\Lambda(\beta)$  is characterized by Ramírez–Rider–Virág [3], the leading right-tail rate agrees with Dumaz–Virág [4], and at  $\beta=2$  the law is Tracy–Widom [5, 6].

The rate constant  $c$  deserves a sharper statement than “a Painlevé II action, numerically 5.44.” The optimal-fluctuation minimizer solves  $-g'' + xg = g^3$  on  $[0, \infty)$ ,  $g(0) = 0$ ,  $g \rightarrow 0$ , and  $c = \int_0^\infty g'^2$ .

**Proposition 2** (The rate-constant identity chain). *With  $g$  the minimizer above,*

$$c = \int_0^\infty g'^2 = \int_0^\infty x g^2 = \frac{1}{2} \int_0^\infty g^4,$$

*i.e.  $A = B = \frac{1}{2}D$  in the notation  $A = \int g'^2$ ,  $B = \int xg^2$ ,  $D = \int g^4$ . Numerically  $c = 5.4438973582$ ;  $c$  has no elementary closed form (inverse-symbolic search against  $\pi$ -powers,  $\Gamma(1/3)^3$ ,  $\zeta(3)$ ,  $\log 2$  to 10 digits is null). [established here]*

*Proof.* Multiplying the Euler–Lagrange equation by  $g$  and integrating by parts gives  $A + B = D$  (identity I). Multiplying by  $xg'$  and integrating by parts gives  $\frac{1}{2}A + \frac{1}{4}D = B$  (identity II, the Pohozaev/dilation identity). Eliminating  $D = A + B$  from (II) yields  $\frac{3}{4}A = \frac{3}{4}B$ , so  $A = B$  and  $D = 2A$ . □

At the process level, the leading escape of a spatially extended folded medium realizes the full Airy<sub>2</sub> process—not merely its TW<sub>2</sub> marginal—if and only if the transverse coupling is disordered, long-range, and stochastically drifting; the increment-variance exponent (Brownian 1 vs. ballistic 2), not the shared marginal, is the discriminant. These process-level statements are numerical; rigorous convergence to the Airy<sub>2</sub> process (Brownian Gibbs / line ensemble) is open (§8).

#### 4. RESURGENT DEPTH AT THE FOLD

The TW<sub>β</sub> right tail carries resurgent structure in *two* independent directions: a *level* direction ( $a \rightarrow \infty$  at fixed  $\beta$ ), governed by Painlevé II and closed by Costin’s theorem [9, 10], and a *noise* direction ( $\beta \rightarrow \infty$  at fixed  $a$ ), whose  $1/\beta$  loop expansion is separately median Borel summable by elementary means [11]. The all-orders soft-tail form is that of Borot–Nadal [7], with the large- $\beta$  reduction to PII from Comtet–Le Doussal–Smith [8]. The level direction rests on an exact linearization: the Jacobi operator  $\mathcal{J} = -\partial_x^2 + 6p_\star^2 - 2(x + a)$  is the Painlevé II linearization  $\mu^2(-\partial_s^2 + 6q^2 + s)$ , the instanton being the  $\alpha = \frac{1}{2}$  Airy special solution of PII [16, 17, 18, 19].

**Theorem 3** (Stokes constant of the backbone; exact). *The  $s \rightarrow -\infty$  Hastings–McLeod coefficients are non-sign-alternating; the Borel singularity sits on the positive axis at action  $A = 2\sqrt{2}/3$ , index  $\gamma = -\frac{1}{2}$ , with Stokes constant*

$$C = \sqrt{\frac{2}{3\pi^3}} = \frac{1}{\pi} \sqrt{\frac{2}{3\pi}} = 0.146632271193848478\dots$$

*matching the Hastings–McLeod value [12, 13, 14] and refuting the natural guess  $1/(4\sqrt{\pi})$ . The  $a$ -dependence is algebraic:  $C$  factorizes as a transverse Kramers determinant  $1/\pi$  times a Gelfand–Yaglom zero-mode Jacobian  $\sqrt{2/(3\pi)} = \|\psi_0\|/\sqrt{2\pi}$  of the tanh-kink escape instanton [15]. [established here] / [proved elsewhere, cited]*

The  $O(1/\beta)$  noise-averaged shift of the connection root admits a closed *form* (though not a closed number).

**Proposition 4** (The connection-root shift  $\Omega_1$ ). *Writing  $\mathbb{E}[\Lambda_0] = \Lambda_0^{\det} + \beta^{-1}\Omega_1 + O(\beta^{-2})$  with  $\Lambda_0 = 2.3381074105$ , second-order Rayleigh–Schrödinger perturbation over the Airy eigenbasis gives  $\Omega_1 = -4 \int_0^\infty \psi_0^2 G_{\text{red}}$ , which reduces to*

$$\Omega_1 = \frac{4\pi}{\text{Ai}'(-\Lambda_0)^2} \left[ \frac{\text{Bi}'(-\Lambda_0)}{\text{Ai}'(-\Lambda_0)} \int_{a_1}^\infty \text{Ai}^4 - \int_{a_1}^\infty \text{Ai}^3 \text{Bi} \right],$$

*the third natural term  $\int_{a_1}^\infty \text{Ai}^3 \text{Ai}'$  vanishing exactly because  $a_1$  is a zero of Ai. Numerically  $\Omega_1 = -1.124813904375434066$  (27 digits). The quartic moment  $\int_{a_1}^\infty \text{Ai}^4$  is provably not boundary-reducible, so  $\Omega_1$  is a genuine Airy-quartic transcendental. [established here]*

**Proposition 5** (Which Painlevé II: saddle vs. distribution). *The leading tail saddle is the  $\alpha = \frac{1}{2}$  Airy special solution; the full  $\beta=2$  distribution is the  $\alpha = 0$  Hastings–McLeod transcendent. These are consistent: the  $\alpha = \frac{1}{2}$  saddle action gives  $\mathcal{T}_\beta(a) \sim a^{-3\beta/4} e^{-(2\beta/3)a^{3/2}}$ , which at  $\beta=2$  reproduces the classical TW<sub>2</sub> right tail  $e^{-(4/3)a^{3/2}}/(16\pi a^{3/2})$  in both exponent and prefactor power (verified: the ratio  $\rightarrow 1$ , e.g. 0.941, 0.957, 0.967 at  $a = 8, 10, 12$ ). Thus  $\alpha = \frac{1}{2}$  governs the saddle/rate and  $\alpha = 0$  the distribution; this is the ordinary saddle-vs-distribution distinction, not a  $\beta$ -dependence. [established here]*

The two directions share the single action  $\Phi(a) = \frac{2}{3}a^{3/2}$ , and the tail is a Borel-summable resurgent trans-series, uniformly in  $a$  for  $a \geq a_0 \approx 1$ .

5. THE CUSP: A PROVABLY NON-INTEGRABLE LAW  $\mathcal{W}$ 

At the cusp the inner equation is the stochastic Weber operator  $u'' = (\text{sign}(Y)Y^2 - \eta\xi)u$ ,  $\beta = 4/\eta^2$ , and the escape law  $\mathcal{W}_\beta$  is a genuinely new object. Two obstructions make this a pair of theorems, not a gap.

**Proposition 6** (No integrable closed form). (i) *The deterministic Weber skeleton is an isomonodromy fixed point [17] (the only relevant deformation flows cusp  $\rightarrow$  fold), so  $\mathcal{W}$  is not the  $\sigma$ -function of a Painlevé transcendent.* (ii)  *$\mathcal{W}$  is a first-passage law of an operator unbounded below (escape occurs down the  $-Y^2$  side), hence not a soft-edge Fredholm determinant. [proved elsewhere, cited]*

In particular the ladder analogy “Airy:PII :: Weber:PIV” does *not* extend to the law:  $\mathcal{W}$  is not a Painlevé IV  $\sigma$ -function. What survives is a closed-form *deterministic connection*. The recessive log-derivative at  $Y = 0$  on the confluent ( $Y > 0$ ) side is  $R(\lambda) = -2\Gamma(\frac{3}{4} - \frac{\lambda}{4})/\Gamma(\frac{1}{4} - \frac{\lambda}{4})$ ; the oscillatory ( $Y < 0$ ) side is the confluent equation rotated by  $e^{-i\pi/4}$  with  $\lambda \rightarrow i\lambda$ , giving  $L(\lambda) = 2e^{-i\pi/4}\Gamma(\frac{3}{4} - \frac{i\lambda}{4})/\Gamma(\frac{1}{4} - \frac{i\lambda}{4})$ .

**Proposition 7** (The cusp resonance root is an exact Gamma-equation root). *The resonance condition  $L(\lambda) = R(\lambda)$  is the explicit equation*

$$e^{-i\pi/4} \frac{\Gamma(\frac{3}{4} - \frac{i\lambda}{4})}{\Gamma(\frac{1}{4} - \frac{i\lambda}{4})} + \frac{\Gamma(\frac{3}{4} - \frac{\lambda}{4})}{\Gamma(\frac{1}{4} - \frac{\lambda}{4})} = 0,$$

*with root  $\lambda_0 = 0.8896052163757951(1 - i)$ . Its phase is exactly  $-\pi/4$ : on the ray  $\lambda = \mu(1 - i)$  the two  $\Gamma$ -arguments are complex conjugates, so the equation reduces to the real condition  $\arg[\Gamma(\frac{3}{4} - \frac{\mu(1-i)}{4})/\Gamma(\frac{1}{4} - \frac{\mu(1-i)}{4})] = 3\pi/8$ . Moreover  $\arg D'(\lambda_0) = -7\pi/8$  exactly. The modulus  $|\lambda_0| = 1.2580918$  has no elementary closed form. [established here]*

The stochastic data are then a resurgent trans-series in  $\eta^2$ . The leading weak-noise variance,  $\text{Var}(Y^*) = \eta^2(v_0 + v_1\eta^2 + \dots)$ , has an exact one-loop form (see §6);  $v_0 = 0.13429234295678519$ , transcendental. The first noise correction to the connection root,  $\mathbb{E}[\lambda] = \lambda_0 + \eta^2\Omega_2 + O(\eta^4)$ , is  $\Omega_2 = -0.451 + 0.352i$  (also transcendental); Proposition 7 makes its prefactor  $-1/D'(\lambda_0)$  exact (the deterministic Weber Stokes normalization is that of Hao [20]). A median Borel resummation of seven exact weak-noise coefficients reconstructs  $\text{Var}$  at  $\beta=2$ —outside the perturbative radius—to about 1%, which is the sense in which the trans-series *is*  $\mathcal{W}$ . [numerical]

## 6. THE LADDER AND ITS TWO DIRECTIONS

Replacing the turning potential by  $V_q(Y) = \text{sign}(Y)|Y|^q$  gives the swept multicritical family  $u'' = (V_q - \eta\xi)u$ . Three structural layers are uniform across it: (1) the deterministic skeleton is a Bessel function of order  $1/(q+2)$ , and the peel-off levels are its zeros (at  $q=1$ , the exact Airy zeros); (2) the exact Freidlin–Wentzell tails are

$$\mathbb{P}(Y^* < -s) \approx \exp\left(-\frac{|s|^{2q+1}}{2(2q+1)\eta^2}\right) = \exp\left(-\frac{\beta|s|^{2q+1}}{8(2q+1)}\right) \quad (\text{left}), \quad \exp(-\tfrac{4}{3}|s|^{3q/2}) \quad (\text{right});$$

(3) the edge law is a resurgent trans-series. To these we add a uniform leading-order statement.

*Remark 8* (The left-tail constant, made  $\eta$ -explicit). Earlier drafts of this section quoted the left tail as  $\exp(-|s|^{2q+1}/[4(2q+1)])$ , with no  $\eta$ . That expression is the  $\beta=2$  specialisation of the law above, not a general- $\eta$  statement, and taken literally it disagrees with the cusp paper’s  $-\log \mathbb{P} \rightarrow s^5/(10\eta^2) = \beta s^5/40$ . The  $\eta$ -explicit form reconciles the two: at  $q = 2$  it gives exactly  $\beta s^5/40$ , and at  $\beta = 2$  it reproduces the earlier numbers. The constant  $1/[2(2q+1)]$  is verified numerically for  $q = 1, \dots, 5$  by boundary-value solution of the Euler–Lagrange system (measured/predicted ratios 0.839, 0.930, 0.981, 0.997, 0.9995, approaching 1 from below since at fixed  $s$  the steeper  $q$  reaches its asymptote sooner). [numerical]

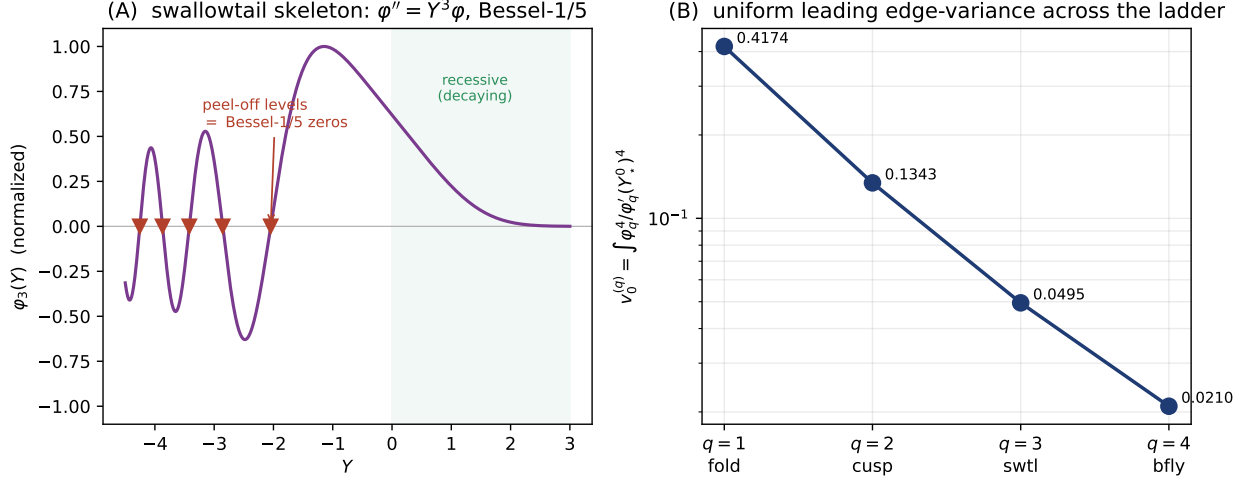


FIGURE 1. The swept ladder. (A) The swallowtail ( $q=3$ ) deterministic skeleton  $\varphi'' = Y^3\varphi$ : recessive (decaying) for  $Y > 0$ , oscillatory for  $Y < 0$ ; its peel-off levels are the zeros of the Bessel-1/5 combination (first five agree to  $10^{-14}$ ). (B) The uniform leading edge-variance  $v_0^{(q)}$  of Proposition 9 across the ladder.

**Proposition 9** (Uniform leading edge-variance). *Let  $\varphi_q$  be the recessive solution of  $\varphi'' = \text{sign}(Y)|Y|^q\varphi$  and  $Y_\star^0$  its first node swept from  $+\infty$ . Then the leading weak-noise variance coefficient is*

$$v_0^{(q)} = \frac{\int_{Y_\star^0}^{\infty} \varphi_q(Y)^4 dY}{\varphi_q'(Y_\star^0)^4}.$$

*The derivation is a first-order node-shift computation in which the Wronskian identity at the node,  $W = -\varphi'(Y_\star^0)\psi(Y_\star^0)$ , cancels the dominant-solution factor. Numerically  $v_0^{(1)} = 0.41743$ ,  $v_0^{(2)} = 0.13429$ ,  $v_0^{(3)} = 0.04953$ ,  $v_0^{(4)} = 0.02097$ ; the  $q=1$  node is the exact first Airy zero. [established here]*

The swallowtail rung ( $q=3$ ,  $A_4$ ) is verified in full: the skeleton is Bessel-1/5 (its first five peel-off levels match the matched Bessel combination—the symmetric  $J_{1/5} + J_{-1/5}$ —to  $10^{-14}$ ), and the standardized escape moments at  $\beta=2$  are skewness  $+0.951$  (predicted  $+0.95$ ) and excess kurtosis  $+0.156$  (predicted  $+0.16$ ), from  $2.8 \times 10^6$  samples. [numerical]

**Swept versus genuine.** This swept ladder must not be confused with the genuine Berry–Upstill diffraction-catastrophe ladder [21]  $u^{(n)} = (Y - \eta\xi)u$ , an  $n$ -th order equation: fold ( $A_2$ ) is Airy, cusp ( $A_3$ ) is *Pearcey* (third order), swallowtail ( $A_4$ ) is the swallowtail function (fourth order). The two families *coincide only at the fold and diverge in opposite directions*. As  $q$  rises the swept laws grow more skewed and sub-Gaussian (skewness  $+0.20, +0.61, +0.95$ ); as the order  $n$  rises the genuine laws trend toward Gaussian (skewness  $+0.225, +0.120, +0.058$ , the first matching  $TW_2$  and validating the integrator). [numerical] The interpretation is structural: the scalar one-fast-variable pipeline of §2 (Cole–Hopf of a scalar Riccati) produces a *second-order* Schrödinger operator, hence the swept ladder; the genuine swallowtail requires a fourth-order linear inner equation, which does not arise from a scalar canard and needs higher-order fast structure. The family therefore does not terminate—it *splits*, and genuine  $A_4$  is the point at which the scalar theory bridges out (Figure 2).

## 7. PHYSICAL REALIZATION, HONESTLY POSITIONED

The turning order is not abstract: in the coupled fold–Hopf–relaxation neuron model the electrical coupling  $g$  sets the turning structure through  $\Delta(g) = 2\sqrt{-2g/3}$ , so tuning  $g$  moves the

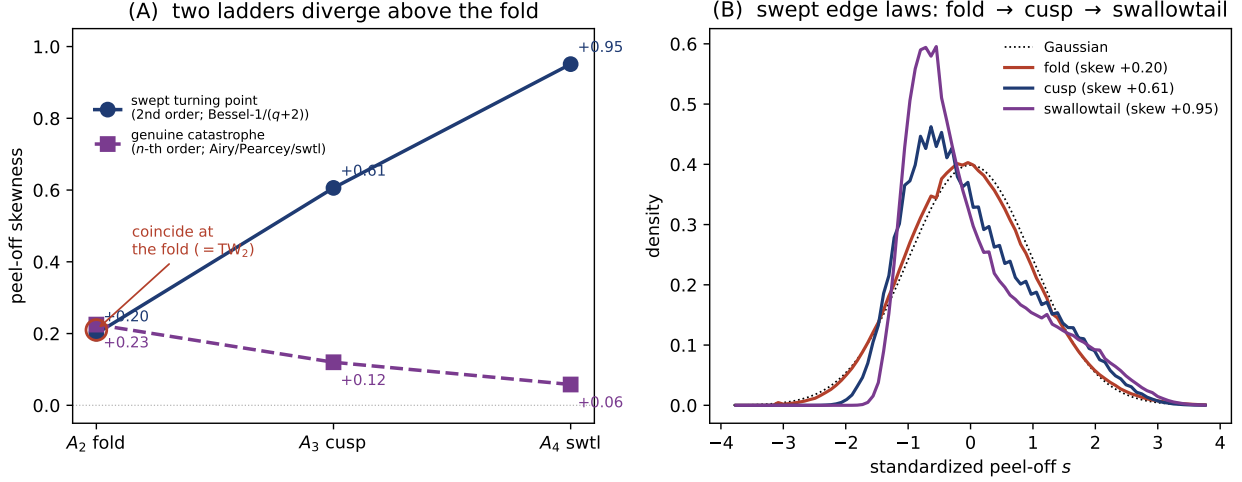


FIGURE 2. The two ladders. (A) Peel-off skewness: the swept turning-point ladder (2nd order) grows more sub-Gaussian with  $q$ , while the genuine Berry–Upstill ladder ( $n$ -th order) trends toward Gaussian; they coincide only at the fold ( $= TW_2$ ). (B) The swept standardized edge laws fold $\rightarrow$ cusp $\rightarrow$ swallowtail, from  $2.8 \times 10^6$  Monte-Carlo escapes each at  $\beta=2$ .

system along the swept ladder—fold (TW) escape statistics at weak coupling, through the cusp (sub-Gaussian, negative excess kurtosis) at  $g_{\text{crit}}$ . The sign of the escape-timing excess kurtosis is the observable rung signature.

We position this carefully. The single most novel content of the program is the *theory spine*: the fold= TW theorem, the resurgent structure with its exact Stokes constant, and the non-integrable cusp law  $\mathcal{W}$ . The *neuron realization*, by contrast, sits squarely within the Berglund–Gentz [22] and Berglund–Gentz–Kuehn [23] slow–fast noise framework; the folded-node / mixed-mode noise phenomenology it touches is their territory, and the clean closed forms found there (e.g. the universal fold-escape prefactor  $\sqrt{4\pi \ln 2}$ ) are consolidations within a known framework, not new phenomena. We claim the demonstration—that the ladder *lives* in the neurons—not novelty in the neuronal modelling.

## 8. THE DEMARCATED OPEN PROGRAM

What closes the book is the account above: the proved spine, the resurgent structure to the depth the present methods reach, the ladder and its separation from the genuine catastrophes, and the physical anchor. What remains is an *explicitly bounded* research program, and every substantial direction in it crosses into a different field.

- *Random matrices / topological recursion*. The all-genus coefficients  $R_m$  of the Borot–Nadal soft-tail formula [7] require the full topological-recursion machinery; this is unreachable from within the present methods.
- *Resurgence proper*. A direct exact-WKB / Voros-symbol resurgence for the random operator itself would treat the white-noise potential without averaging; no exact-WKB framework exists for a nowhere-differentiable potential (the Voros symbols  $\int \sqrt{Q}$  are ill-defined path-wise). The scoped attack is parametric-Stokes theory with the noise coupling as deformation parameter.
- *Integrable probability / KPZ*. Process-level convergence to the Airy<sub>2</sub> (and, conjecturally, Pearcey) process needs a Brownian-Gibbs / line-ensemble property for the folded-medium edge; only the marginals and the increment-variance exponent are in hand.



- *Computational neuroscience.* The realization in actual networks and data.

Two residuals are softer and internal: the rough-path rigidity of the rough-potential isomonodromy (the  $O(\varepsilon^2)$  location shift is  $\delta(0)$ -divergent but has a finite Wick-ordered form), and the nonlinear first-passage sensitivity bound  $R_{\text{nl}}$  that stands between the proved sub-Gaussian inner moments and a fully closed connection-phase cumulant bound. Neither is a new framework; both are writeable with effort. Everything past them is a deliberate crossing, chosen on its own merits, not a deficiency of the account here.

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